PRO

- Desiccant equipment tends to be very durable.
- Often this is the most economical means to dehumidify below a 40°F (4.3°C) dew point.
- It eliminates condensate in the airstream, in turn, limiting the opportunity for mold growth.

CON

- Desiccant usually must be replaced, replenished, or reconditioned every 5 to 10 years.
- In comfort applications, simultaneous heating and cooling may be required.
- The process is not especially intuitive, and the controls are relatively complicated.

KEY ELEMENTS OF COST

The following provides a possible breakdown of the various cost elements that might differentiate a building with a rotary desiccant dehumidification system from one without and an indication of whether the net cost is likely to be lower (L), higher (H), or the same (S).

First Cost

• Equipment costs	H
• Regeneration (heat source and supply)	H
• Ductwork	S
• Controls	H
• Design fees	S

Recurring Cost

• Overall energy cost	S/H
• Maintenance of system	H
• Training of building operators	H
• Filters	H

SOURCE OF FURTHER INFORMATION

ASHRAE. 2020. *ASHRAE Handbook—HVAC Systems and Equipment*. Peachtree Corners, GA: ASHRAE.

ASHRAE GreenTip #9-13

Indirect Evaporative Cooling

GENERAL DESCRIPTION

Evaporative cooling of supply air can be used to reduce the amount of energy consumed by mechanical cooling equipment. Two general types of evaporative cooling—direct and indirect—are available. The effectiveness of either of these methods is directly dependent on the extent that dry-bulb temperature exceeds wet-bulb temperature in the supply airstream.

Direct evaporative cooling introduces water directly into the supply airstream, usually with a spray or wetted media. As the water absorbs heat from the air, it evaporates. While this process lowers the dry-bulb temperature of the supply airstream, it also increases the air moisture content.

Two forms of indirect evaporative cooling (IEC) are described below.

1. **Coil/cooling Tower IEC.** This type uses an additional water-side coil to lower supply air temperature. The added coil is placed ahead of the conventional cooling coil in the supply airstream and is piped to a cooling tower where the evaporative process occurs. Because evaporation occurs elsewhere, this method of precooling does not add moisture to the supply air, but it is somewhat less effective than direct evaporative cooling. A conventional cooling coil provides any additional cooling required.
2. **Plate Heat Exchanger IEC.** This is composed of sets of parallel plates arranged into two sets of passages separated from each other. In a typical arrangement, exhaust air from a building is passed through one set of passages, during which it is wetted by water sprays. A stream of outdoor air is coincidently passed through the other set of passages and is cooled by heat transfer through the plates by the wetted exhaust air before being supplied to the building. Alternatively, the exhaust air may be replaced by a second stream of outdoor air. The wetted air is reduced in dry-bulb temperature to be close to its wet-bulb temperature. The stream of dry air is cooled to be close to the dry-bulb temperature of the wetted exhaust air. In some applications, the cooled stream

of outdoor air is passed through the coil of a direct expansion refrigeration unit, where it is further cooled before being supplied to the building. This system is an efficient way for an all outdoor air supply system. The plates in the heat exchanger can be formed from various metals and polymers. Consideration needs to be given to preventing the plate material from corroding.

WHEN/WHERE IT'S APPLICABLE

This may be used in climates with low wet-bulb temperatures where significant amounts of cooling are available. In such climates, the size of the conventional cooling system can also be reduced.

In more humid climates, indirect evaporative cooling can be applied during nonpeak seasons. It is especially applicable for loads that operate 24 h/day for many days of the year.

PRO

- Indirect evaporative cooling can reduce the size of the conventional cooling system.
- It reduces cooling costs during periods of low wet-bulb temperature.
- It does not add moisture to the supply airstream (in contrast, direct evaporative cooling does add moisture).
- It may be designed into equipment such as self-contained units.
- There is no cooling tower or condenser piping in the plate heat exchanger IEC that is described.

CON

- Air-side pressure drop (typically 0.2 to 0.4 in. of water [50 to 100 Pa]) increases due to an additional coil in the airstream.
- To make water cooler in the coil/cooling tower IEC, the cooling tower fans operate for longer periods of time and consume more energy.
- For the coil/cooling tower IEC, condenser piping and controls must be accounted for during the design process.

KEY ELEMENTS OF COST

The following provides a possible breakdown of the various cost elements that might differentiate an IEC system from a conventional one and an indication of whether the net cost for the hybrid option is likely to be lower (L), higher (H), or the same (S). This assessment is only a perception of what might be likely, but it obviously may not be correct in all situations. There is no substitute for a detailed cost analysis as part of the design process. The listings below may also provide some assistance in identifying the cost elements involved.

First Cost

- Indirect cooling coil H
- Decreased conventional cooling system capacity L
- Condenser piping, valves, and control H

Recurring Cost

- Cooling system operating cost L
- Supply fan operating cost H
- Tower fan operating cost H
- Maintenance of indirect coil S

SOURCES OF FURTHER INFORMATION

ASHRAE. 2019. Chapter 52, "Evaporative Cooling." *ASHRAE Handbook—HVAC Applications*. Peachtree Corners, GA: ASHRAE.

ASHRAE. 2020. Chapter 41, "Evaporative Air-Cooling Equipment." *ASHRAE Handbook—HVAC Systems and Equipment*. Peachtree Corners, GA: ASHRAE.

ASHRAE GreenTip #9-14

Thermally Actuated Diffusers

GENERAL DESCRIPTION

Conventional air distribution systems typically group several rooms into a typical VAV box or zone that is controlled by a single thermostat. Because rooms within the zone can have different heating/cooling requirements, the system's response to heating/cooling loads within the zone may not be consistent with load requirements. The result can be a lack of thermal comfort of the occupants in each room by providing rather all cooling or all heating irrespective of the occupants need. In addition, the re-heat at the VAV box results in increase energy consumption.

Thermally actuated diffusers commonly use thermal actuators to modulate the flow rate of supply air into a room according to a desired temperature setting. The diffuser works independent of a BAS and does not require external power since the close/open action is cause by a thermal gel that is purely mechanical. The diffuser measures the room temperature by inducing room air into the diffuser core and passing it over a thermal actuator. Linkage contained within the core adjusts the damper based on the difference between room and desired set point temperatures. Some advanced models can be connected to the BMS systems.

WHEN/WHERE IT'S APPLICABLE

Thermally actuated diffusers can be uised in any all-air systems and are generally found economically feasible when compared to a VAV box is serving less than five air outlets.

PRO

- Improved thermal comfort
- Reduced energy consumption
- With proper zoning, possibility of elimination of the heating hot water system including the boiler.

CON

- Higher first cost
- Required an engineered duct design
- Basic models cannot connect to the BMS

KEY ELEMENTS OF COST

The following provides a possible breakdown of the various cost elements that might differentiate a building with a rotary desiccant dehumidification system from one without and an indication of whether the net cost is likely to be lower (L), higher (H), or the same (S).

First Cost

• Equipment costs	H
• Ductwork	S
• Controls	S
• Design fees	S

Recurring Cost

• Overall energy cost	L
• Maintenance of system	L
• Training of building operators	S
• Filters	S

SOURCE OF FURTHER INFORMATION

Engineering Guide VAV diffusers. Price Industries. https://www.priceindustries.com/content/uploads/assets/literature/engineering-guides/vav-diffusers-engineering-guide.pdf.

ASHRAE GreenTip #9-15

Automated Refrigerant Leak Detection and Recovery

GENERAL DESCRIPTION

In the past, leaking chillers were acceptable because of the ready supply and the low cost of refrigerant. Today if a refrigerant leaks in chillers (be it R134a, R410a, HFOs), it not only cost the building owner but also the refrigerant released to the atmosphere will cause global warming, increase carbon footprint and in a way affect climate change. This may not be acceptable under the US Clean Air Act.

Automated refrigerant leakage detection and recovery system not only prevent further leakage of refrigerant being dispersed into the atmosphere but also prevent wastage of refrigerant, achieve cost savings, minimize risk and protect the safety of mechanical room personnel and the environment. If owners choose to follow any green building certification, automated refrigerant leakage detection can benefit them not just by being environmentally sound but also by lowering operation cost, reducing carbon footprint, reducing liability and risk, enhancing marketability through higher return on investment and increasing tenant rental.

WHEN/WHERE IT'S APPLICABLE

Automated refrigerant leakage detection and recovery systems are typically used in mechanical rooms where chillers are installed. Some chillers lose 2% to 15% of their refrigerant charge annually through different mechanisms. In the past, all of the lost refrigerant was released to the atmosphere.

Refrigerant loss occurs primarily through four different channels: leaks, poor service procedures, contamination, and purge cycles. Depending on the local service standards, the relative magnitude of these channels can vary widely.

The case of high-pressure chillers is slightly different than low pressure chillers. High pressure chillers maintain positive pressure throughout the refrigerant circuit; therefore, they do not require purge units. Aside from this difference, the refrigerant loss mechanisms are basically the same as in low pressure chillers. All of the losses for both high- and low-pressure chillers can be significantly reduced by using the auto leakage detection and recovery system installed in the chillers.

The largest source of refrigerant loss from chillers is equipment leakage caused by normal operation and wear. Leaks are most often found in tubing, flanges, O-rings, seals and other connections. Losses from equipment leakage can be greatly reduced if leaks are identified and repaired, though in recent times, leak rate of new chillers are greatly reduced.

DETAILED DESCRIPTION

A complete automated refrigerant leak detection and recovery system will generally consist of:

- nondispersive infrared (NDIR) refrigerant sensors with pump aspirated sampling system
- refrigerant recovery unit
- refrigerant gas cylinders
- steel platform weighing scales to determine amount of refrigerant recovered
- refrigerant ball valves
- electrical enclosure with general alarm
- copper piping and wiring

The refrigerant leakage and recovery system works by constantly monitoring for any refrigerant leak. This is achieved by NDIR sensor with pump aspirated sampling system (per chiller) which is capable of picking up leaks better than a static sensor. Plus, the use of superior NDIR technology ensures there is minimal false leak detection from other VOC's and fumes.

The design for the project will depends on the number of chillers and type of refrigerant used, refrigerant charge and blend of the largest chiller, layout, air circulation, and leakage sensitivity among other key parameters.

Whenever there is a chiller leak, the refrigerant being heavier than air tends to slowly settle down at ground level. Because pf that, the refrigerant sensor (based on the particular refrigerant used in the chillers) is generally installed 8 in. (200 mm) above the ground near both ends of every chiller.

In the event of a leak, the gas detection system or fuzzy logic code in the chiller programmable logic controller (PLC) detects a possible leak in progress and a signal will be sent to the Refrigerant Recovery Unit, then a signal in the form of a dry contact closure will be sent to the chiller PLC. Once the chiller has readied itself for refrigerant recovery it sends a signal back confirming that refrigerant recovery may proceed using a high-speed vacuum pump in the refrigerant recovery unit. The activation signal may come from any sensor source. The HVAC engineers, architects along with the BMS engineers will ultimately decide which configuration best suits their needs.

The system will allow refrigerant to flow from the chiller to the refrigerant storage vessels for one minute before its pumping unit begins. The system will recover the refrigerant until a nominal pressure of around 3 to 4 psig (25kPa) is achieved. This value may be altered, however it is desirable to keep this pressure above atmospheric pressure as air maybe recovered with the refrigerant; which is undesirable.

The system will remain active until it is reset or switched off. This strategy has been chosen in the event liquid refrigerant remains "trapped" in the chiller after the system has recovered to a nominal pressure of 25kPa. At this pressure during recovery, any liquid refrigerant may likely be very cold. For example, for R134a it is likely to be around –5°F to –10°F (roughly –22 °C). Thus, it may be some time before its temperature/pressure increases and therefore the recovery system will need to remain active.

Integral to this system is the measurement of the recovered refrigerant. Each recovered refrigerant has a measuring scale so the recovered refrigerant can be logged and the loss of refrigerant calculated. On the charging cycle, this system accurately measures the charge into the chiller allowing any initial refrigerant loss to be added and logged.

PRO

- Reduce leak refrigerant emission into the atmosphere thereby reducing contribution to greenhouse gas emissions
- Potential for reduced owner operating cost
- Save liabilities and minimize personnel risk working in mechanical room